Formal regularity of the conformal field equations with Dirac, Higgs and gauge matter: computations

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Abstract

This note collects the computations behind Remark 2 of the companion paper. We use Friedrich's conformal field equations [1] and his notion of formal regularity, and treat in turn: the coupled Einstein–Dirac system (the geometric equations are formally regular in kinetic form; the singularity is confined to the matter equation); the Higgs sector with Yukawa and gauge-boson masses (the fermion mass coefficient is regular exactly when the rescaled Higgs is, and the W, Z masses are regular when it is); the coupled Einstein–Yang–Mills–Higgs–Dirac system (formally regular if and only if $m_h^2 + 4\lambda\xi = \frac{2}{3}\lambda$); and a test Dirac field with the dust weight of Friedrich's treatment of pressure-free matter [2] (no pole in the source of the rescaled Weyl tensor; a power series for single-exponent solutions, zero for $\mathbf{k} = 0$, bounded without limit for solutions with both exponents and $\mathbf{k} \neq 0$). All statements were obtained with symbolic or numerical scripts and audited by two isolated AI agents each (the corrected script of the dust-weight section was not re-audited); none has been read by a human. They concern formal regularity, not existence or stability.

1 Conventions

In this note we follow Friedrich [1, 2]: the physical metric is $\hat{g}$, the conformal metric $g = \Omega^2\hat{g}$, signature $(-,+,+,+)$, cosmological constant $\lambda > 0$, κ the gravitational coupling (the companion paper puts the hat on the unphysical side). The unknowns of the conformal field equations are Ω, $s = \frac{1}{4}\Box\Omega + \frac{1}{24}\Omega R[g]$, the Schouten tensor $L_{\mu\nu}$ of g, the rescaled Weyl tensor $W = \Omega^{-1}C[g]$ and the frame and connection, with the equations (2.13)–(2.17) of [1]. The equation for W reads

$$\nabla_\mu W^\mu{}_{\rho\nu\lambda} = 2\,\Omega^{-1}\,\hat{\nabla}_{[\nu}\hat{L}_{\lambda]\rho}, \qquad \hat{L}_{ab} = \tfrac{\kappa}{2}\big(\hat{T}_{ab} - \tfrac{1}{3}\hat{T}\hat{g}_{ab}\big) + \tfrac{\lambda}{6}\hat{g}_{ab}. \tag{1}$$

Formal regularity (Friedrich's Theorem 3.1): the right-hand sides, written as functions of the rescaled unknowns and of their derivatives treated as independent variables (jets at a point), depend smoothly on Ω up to $\Omega = 0$, as in Definition 1(c) of the paper; in particular they contain no negative powers of Ω, no $\ln\Omega$ and no factor $\exp(\mathrm{i}c\ln\Omega)$, $c \neq 0$, multiplying a non-zero coefficient. The second and third derivatives of Ω are eliminated with (2.14) and its derivative, ∇s with (2.15) (whose matter term carries the factor Ω^{-1}), and $|\mathrm{d}\Omega|^2$ with $|\mathrm{d}\Omega|^2 = -\lambda/3 + 2\Omega s + \frac{\kappa}{12}\hat{T}$ [1, (3.8)]. Calibration: with these rules, the scalar case reproduces exactly the coefficient $\frac{\kappa}{3}(m^2 - \frac{2}{3}\lambda)$ of Friedrich's Theorem 3.1, in both directions, and the conformally coupled massless scalar gives no singular coefficient.

2 Einstein–Dirac

With $\psi = \Omega^{3/2}\psi_r$ and the symmetrised kinetic tensor $T_{\mu\nu} = \frac{\mathrm{i}}{4}\big(\bar\psi\gamma_{(\mu}\nabla_{\nu)}\psi - \nabla_{(\nu}\bar\psi\gamma_{\mu)}\psi\big)$, which contains no mass, one has, identically off shell and algebraically in ψ_r and its first derivatives,

$$\hat T_{\mu\nu}[\psi;\hat g] = \Omega^2\, T_{\mu\nu}[\psi_r; g]. \tag{2}$$

Hence $\hat T = \Omega^4\Theta_{\mathrm{tr}}$, the trace-free part is $\Omega^2(\cdots)$, $\hat L = \Omega^2 Y + \frac{\lambda}{6}\hat g$, and the lowest power of Ω in the right-hand side of (1) is 0 for generic smooth Y. The geometric equations are formally regular for every mass. The mass enters only through the matter equation, $(\mathrm{i}\gamma\cdot\partial - m/\Omega)\psi_r = 0$ on a flat conformal background, i.e. through the coefficient m/Ω and, for the derivatives of ψ_r among the unknowns, through its derivatives. With the stress tensor written as "kinetic $-\hat g\,\mathcal{L}$", the Lagrangian term $\mathcal{L} = \Omega^4\bar\psi_r(\mathrm{i}\gamma\cdot\partial - m/\Omega)\psi_r$ gives terms of order Ω^{-1} proportional to m; they vanish modulo the Dirac equation. Conclusion: for the unknown ψ_r, the coupled system is formally regular if and only if $m = \Omega\mu$ with μ smooth; on solutions the singularity appears as the oscillation $\Omega^{\pm\mathrm{i}m/H}$. Script `ed_check.py`; audit: concordance AU-V2 (the connection check of the script was circular and was replaced by an independent determination of the spin connection by the blind auditor).

3 Higgs, Yukawa and gauge-boson masses

With $H = \Omega H_r$, $\psi = \Omega^{3/2}\psi_r$ and A of weight 0, the terms

$$\sqrt{-\hat g}\, y\, \bar\psi_L H \psi_R + \text{h.c.} \quad \text{and} \quad \sqrt{-\hat g}\,\hat g^{mn}|H|^2 A_m A_n$$

have weight 0: Yukawa and gauge-boson mass terms are conformally invariant. The rescaled Dirac equation carries the coefficient $yH_r = m_f/\Omega$, and the rescaled W, Z mass is $g|H_r|$. By contrast $\sqrt{-\hat g}\, m_h^2|H|^2$ has weight -2: it is the term $\Omega^{-2}m_h^2|H_r|^2$, compensated by the curvature term exactly at Friedrich's condition. For a homogeneous Higgs near the symmetric point on de Sitter space with Hubble rate H_Λ ($\lambda = 3H_\Lambda^2$), with $M_2 = (m_h^2 + 12\xi H_\Lambda^2)/H_\Lambda^2$, the indicial equation is $\alpha(\alpha-3) + M_2 = 0$ and, for generic data,

M_2	Dirac field continuous	energy density bounded	coefficient smooth
$M_2 \le 0$	no	no	no
$0 < M_2 < 2$	yes	no	no
$M_2 = 2$	yes	yes	yes
$2 < M_2$	yes	yes	no

($M_2 = 9/4$ gives a logarithm, $M_2 > 9/4$ an oscillation $\Omega^{1/2\mp\mathrm{i}\nu}$.) The three columns are conditions (a), (b), (c) of Proposition 4 of the paper. $M_2 = 2$ is sufficient, and necessary for generic data; non-generic data with a vanishing leading coefficient have other exponents. Script `hx_check.py`; audit AU-V2 (the "if and only if" was restricted to generic data; the W, Z statement to sufficiency; the symmetric-point hypothesis made explicit).

4 Einstein–Yang–Mills–Higgs–Dirac

Unknowns A_μ, $H_r = \Omega^{-1}H$ (complex doublet, potential $m_h^2|H|^2 + \lambda_h|H|^4$, coupling $\xi\hat R|H|^2$), ψ_r with Yukawa mass. Results:

- Yang–Mills is conformally invariant in four dimensions; the Higgs and Dirac currents are of order Ω^2 and regular.

- Rescaled Higgs equation: $\Box_D H_r = \Omega^{-2}\big(m_h^2+4\lambda\xi-\frac{2}{3}\lambda\big)H_r-(\xi-\frac{1}{6})\kappa\,\Omega^{-2}\hat{T}\,H_r+2\lambda_h|H_r|^2H_r+$ (Yukawa), with $\Omega^{-2}\hat{T}$ regular. A conformal weight $w \neq 1$ for the scalar produces an additional Ω^{-1} term; a non-polynomial invariant such as $|H|^3$ would also produce one.

- Source of (1): lowest power Ω^{-2} with coefficient exactly $\frac{\kappa}{3}\big(m_h^2+4\lambda\xi-\frac{2}{3}\lambda\big)\sum_i \phi_i^2\,\nabla_{[\nu}\Omega\,g_{\lambda]\rho}$, and an Ω^{-1} coefficient $\frac{\kappa}{6}\big(m_h^2+4\lambda\xi-\frac{2}{3}\lambda\big)\nabla_{[\nu}(\sum_i \phi_i^2)g_{\lambda]\rho}$ ($H_r = (\phi_1+\mathrm{i}\phi_2)/\sqrt{2}$ per complex component); both vanish together; no Ω^{-3}. Neither A, nor $\mathrm{d}A$, nor λ_h, nor the Yukawa coupling appears in a singular coefficient. The ξ contribution comes as $-\frac{2}{3}$ from $\hat{G}f$ and $+2$ from the improvement term.

- With the Yukawa mass, the Lagrangian term of the Dirac stress tensor is of order Ω^2 in $\hat{L}$, so the representation dependence of Section 2 disappears.

Conclusion: formally regular if and only if $m_h^2 + 4\lambda\xi = \frac{2}{3}\lambda$ (for $\xi = 0$, Friedrich's condition). Script `fw3_check.py`; audit AU-V2: the blind auditor rederived the condition and the Ω^{-1} coefficient; the adversarial auditor found a transcription error in the script (factor Ω instead of Ω^{-1} in (2.15)), whose correction leaves the output unchanged byte for byte. The non-abelian case is covered by weight counting only.

5 Test Dirac field with the dust weight

Identity. For any symmetric $\hat{T}$, set $\tau_{jk} = \Omega^{-3}\hat{T}(\hat{e}_j, \hat{e}_k)$ in a $\hat{g}$-orthonormal frame $\hat{e}_j = \Omega e_j$ (Friedrich's weight for dust, $\rho = \Omega^{-3}\hat{\rho}$), $\tau' = \tau - \frac{1}{3}(\mathrm{tr}\,\tau)g$, $n = \mathrm{d}\Omega$. Then, with $\kappa = 1$,

$$2\Omega^{-1}\hat{\nabla}_{[k}\hat{L}_{l]j} = \tfrac{1}{2}\big(\nabla_k\tau'_{lj} - \nabla_l\tau'_{kj}\big) + \Omega^{-1}S_{jkl}, \tag{3}$$

$$S_{jkl} = \tfrac{1}{2}\big[2(n_k\tau'_{lj} - n_l\tau'_{kj}) - g_{kj}n^m\tau'_{lm} + g_{lj}n^m\tau'_{km}\big]. \tag{4}$$

Checked exactly (sympy) with a flat conformal metric and generic polynomial Ω and $\hat{T}$; confirmed by an auditor for arbitrary functions and for a curved conformal metric. For dust, $S(\rho UU, n) = \rho Z$ with Friedrich's Z_{jkl} of [2, (3.20)]. For timelike n, $S = 0$ if and only if $\tau = \rho\,\nu\nu$ with ν the unit normal along n (rank 9 on the 10 components): only dust at rest along $\mathrm{d}\Omega$ escapes the Ω^{-1} term.

Dirac. Notation as in Remark 1 of the paper: $\chi_0^{\pm}$ are the components of the limit spinor in the eigenspaces $\beta = \pm 1$, and single-exponent solutions are those with $\chi_0^- = 0$ or $\chi_0^+ = 0$. On de Sitter space ($\Omega = -H\eta$), for a mode $\psi = a^{-3/2}u\,e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}}$ with $B = u^\dagger\beta u$, $N = u^\dagger u$, $J_i = u^\dagger\alpha_i u$:

$$\tau_{00} = mB + \Omega\,\mathbf{k}\cdot\mathbf{J}, \qquad \tau_{0a} = -\Omega k_a N, \qquad \tau_{ab} = \tfrac{\Omega}{2}(k_aJ_b + k_bJ_a). \tag{5}$$

The spin-connection terms cancel in the symmetric part (derived independently by the two auditors of this computation, RA-1 and RA-2); the mass cancels pointwise from τ_{0a} and τ_{ab} ($\psi^\dagger\{\alpha_i, \beta\}\psi = 0$), so these are $O(\Omega)$ for every solution, wave packets included. With the Fuchsian form $u = \Omega^{\mathrm{i}m/H}P_+ + \Omega^{-\mathrm{i}m/H}P_-$ ($P_\pm$ entire):

solution	τ	source of (1)
both exponents, $\mathbf{k} \neq 0$	bounded; factor $\Omega^{\mp 2\mathrm{i}m/H}$	no pole; bounded, no limit
single exponent (incl. packets of one type)	power series in Ω	power series in Ω
$\mathbf{k} = 0$	$\tau = mB\,e_0e_0$, constant	0
$m = 0$	power series	power series

The density limit $mB \to m(\|\chi_0^+\|^2 - \|\chi_0^-\|^2)$ has no fixed sign, unlike Friedrich's $\hat{\rho} \geq 0$. The other conformal field equations, (3.3)–(3.6) of [2] ((3.7) being the Weyl equation treated above), have no negative power of Ω in this weight (established by an auditor). The Dirac equation itself

keeps a regular singular point (pole m/Ω for u, bounded phase coefficient $\boldsymbol{\alpha}\cdot\mathbf{k}\,e^{2\mathrm{i}\beta(m/H)\ln\Omega}$ for $\chi = e^{-\mathrm{i}\beta(m/H)\ln\Omega}u$); no transformation $u = V\chi$ with V smooth, or independent of $\mathbf{k}$, makes its coefficients smooth. The current of a generic solution is not asymptotically at rest (J/N of order one and oscillating), so the transposition of Friedrich's flow-adapted gauge is not straightforward; it has not been attempted. Scripts `rj_check.py`, `rj_check_v2.py` (sign of τ_{0a} corrected, single-exponent class tested); audit: the first version described single-exponent solutions as only continuous; both auditors showed they are analytic. Limits: test field on exact de Sitter space (first stage); the coupled system in the dust weight, which could reintroduce poles and affect the single-exponent class, FLRW backgrounds tending to de Sitter, and inhomogeneous data are not treated.

6 Scripts and audit status

computation	script (sha256, 16 hex)	audit
calibration (Friedrich scalar)	`cal_f.py` (`47fb1d46aae577d4`)	two AI auditors
Einstein–Dirac	`ed_check.py` (`216d24c436cca21d`)	AU-V2
Higgs, Yukawa, W, Z	`hx_check.py` (`715b23948563b3b2`)	AU-V2
Einstein–Yang–Mills–Higgs–Dirac	`fw3_check.py` (`cab4bf099d588968`)	AU-V2
dust weight (test field)	`rj_check_v2.py` (`bda4f1c2ba1533fc`)	AU-V3 partial; v2 corrected, not re-audited

All scripts, outputs, audit reports and concordance notes are archived in the programme repository (https://github.com/droppy94/LC_Raccord, branch `archive/notes-2026-10`). "Audit" means two isolated AI agents (one blind derivation, one adversarial reading) and a written concordance; it is not a human review.

References

[1] H. Friedrich, "Smooth non-zero rest-mass evolution across time-like infinity", Ann. Henri Poincaré 16 (2015) 2215, arXiv:1311.0700.

[2] H. Friedrich, "Sharp asymptotics for Einstein-λ-dust flows", Commun. Math. Phys. 350 (2017) 803, arXiv:1601.04506.